

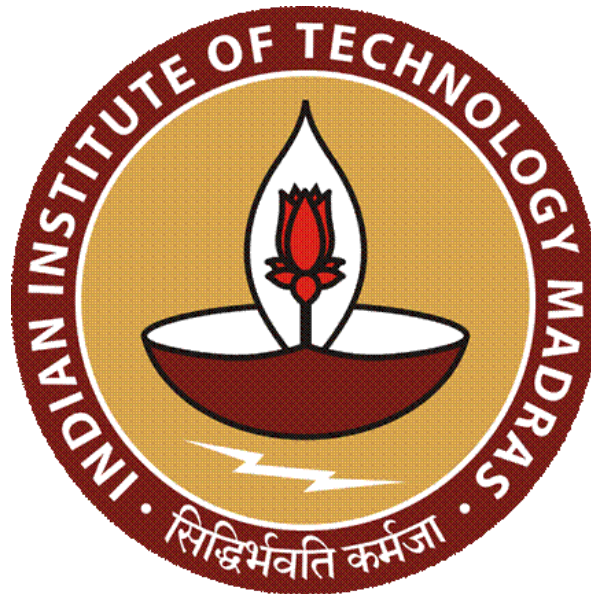
Leveraging Data for Smart Manufacturing: Reducing Costs & Maximizing Sticker Sales

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled “Leveraging Data for Smart Manufacturing: Reducing Costs & Maximizing Sticker Sales”. I extend my appreciation to Navkar Enterprises, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



Signature of Candidate

Name: Vinshi Jain

Date: 1 February 2025

Executive Summary

Navkar Enterprises is a B2B manufacturing unit based in Tri Nagar, Delhi, India. It specializes in footwear stickers and provides customized and mass-produced stickers to shoe manufacturers, wholesalers, and retailers in various markets.

Several of the issues related to inventory management and operational efficiency challenge the company. Excess inventory brings higher storage costs and inaccurate demand forecasting that brings about overproduction. The major objectives include adopting an efficient way of inventory management, accurately measuring and recording inventory and finding ways to cut down costs that would be effective in making the business efficient and increase the profit margins. Using tools such as Excel and Python the business will make better processing of data and making of decisions which eventually result in optimization of stock levels and minimize wastage.

The expected outcome is improved operational efficiency, reduced material wastage, lower production costs, and improved business profitability through optimized inventory management and demand forecasting.

Organization Background

Navkar Enterprises is a footwear sticker manufacturing concern established in the year 2011 by Mr. Rajesh Jain. Its location is 488/34, Onkar Nagar-C, Tri Nagar, New Delhi.

It is a B2B business model which is specialised in producing customized and mass-produced footwear stickers like trendy patterns, cartoon themed stickers, brand logo prints and custom designs to meet client specific requirements.

It employs around 10 people excluding the proprietor Mr. Rajesh Jain. Over the years it has worked to expand its business along different regions in India. The company lays a strong emphasis on their silicone product's quality so that it doesn't get damaged due to wear and tear, moisture and environmental factors.



SILICONE FOOTWEAR STICKERS SAMPLE PHOTOGRAPHS

Problem Statement

After going through discussion with Mr. Rajesh Jain and analysing the data, I observed some of the problems that the business is facing. These are as follows:-

1. Rising competition and Peer Pressure:

They are facing tough competition with the other manufacturing businesses in setting the optimal prices for the stickers which results into price war.

2. Lack of clarity in stock levels and Usage:

Many problems are of over production and stockouts due to lack of real time stock monitoring. Some materials get used up quickly while others sit in storage.

3. Struggles with Inventory Management:

Keeping tracks of raw materials like silicone, vinyl, adhesive and inks is a cumbersome task. There is always a risk of overstocking which may lead to wastage.

Background Of The Problem

Major cause of problems

Navkar enterprises is facing competition where multiple manufacturers are offering similar footwear stickers at low prices due to which the company faces constant pressure to reduce the sticker prices that eventually leads to lower profit margins. Also, the large scale manufacturers have better access to the raw materials at a cheaper price which makes it difficult for the small players to compete.

Overproduction of stickers also leads to shortage leading to excess stock and wastage due to no real time stock monitoring system and fluctuating demand from B2B clients. Some designs become less popular, making it harder to predict inventory needs. Due to improper classification, materials sit unused in the storage.

Raw material price fluctuations lead to higher cost and poor budgeting. Manual inventory counting increases errors and delays in restocking decisions. Inconsistent order patterns also make it difficult to track the stock levels. Shortages of essential materials result in discontinued production, causing missed deadlines and dissatisfied customers.

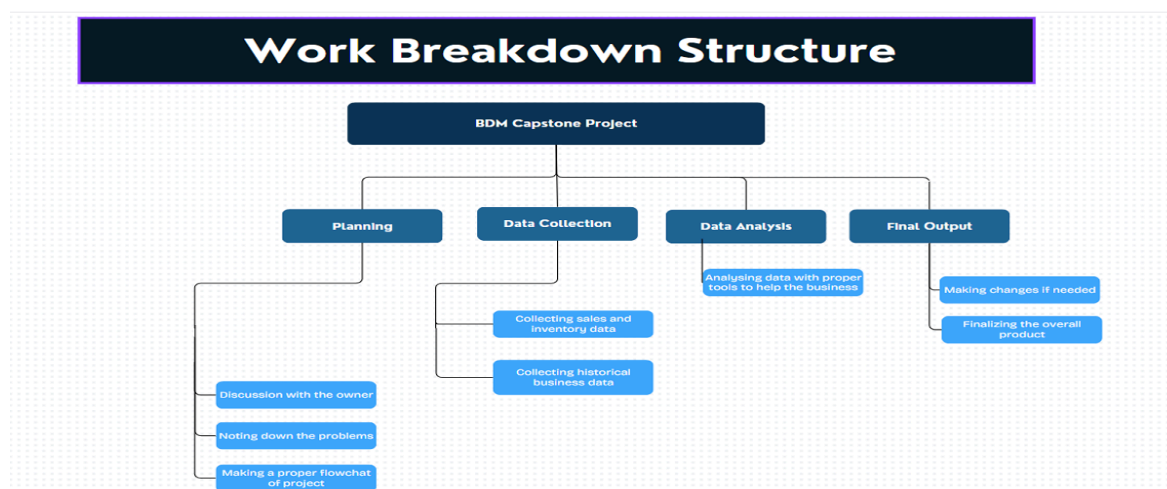
Problem Solving Approach

For the problem of tough competition with other local manufacturers, we can keep a note of what category is getting purchased the most by the customers. This can be done using pareto analysis as that will tell which 20% of the categories of products are generating about 80% of revenue.

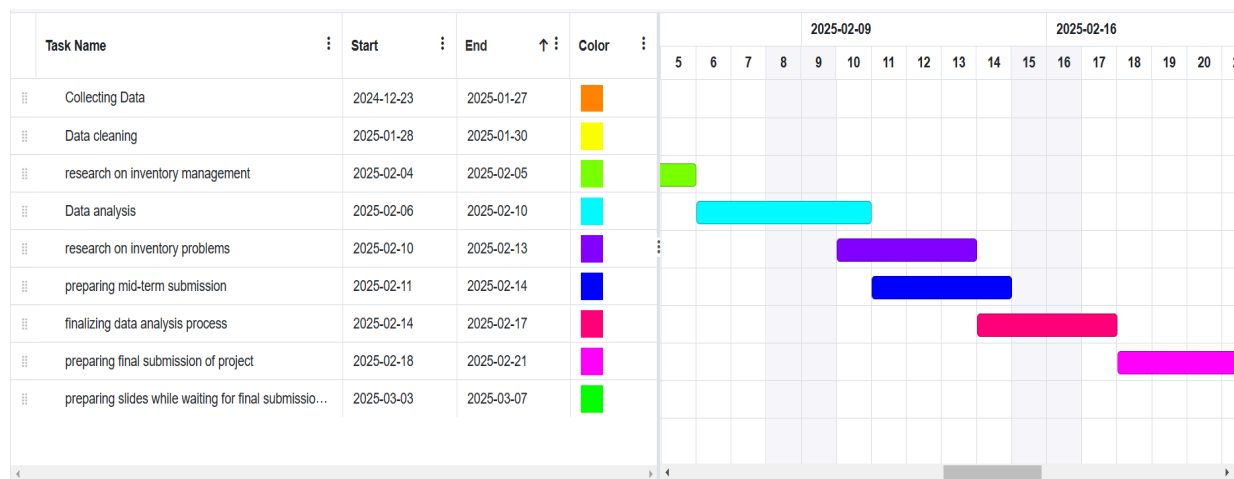
- For the problem of not being able to cope up with the demand during the season, we can use what-if analysis functions of Excel to predict what would be the sales and demand in the next financial year based on the current data. The tool will give a more accurate prediction than oral prediction by the person.
- For the problem of pricing war from competitors, we can optimize the profit keeping the price competitive using optimisation tools in excel.
- For the problem of fluctuating raw materials, we can update and reduce the finished products of certain categories that are generating low revenue and taking high ingredients usage. Even the ingredient can also be replaced with some alternative which nearly keeps the quality of products the same.

Tools for Analysis- The basic tool that will be used for this analysis will be MS Excel. Firstly, the data had been converted from Tally to .xlsx format, because after that only further analysis can be done. There are different functions available in Excel such as Pivot table, Charts and VLOOKUPs which will be employed to solve the problems by analysing the data in structured format. Bar charts will be used to see in which month the revenue is the maximum and in which month it is the minimum. Some other functions like what-if analysis will be used to see what will be the effect of changing some of the values and later to observe the trends.

Expected Timeline



The Project is broken down into the following parts as mentioned in the above Work Breakdown Structure diagram.



The project is anticipated to be completed within the projected time frame, aligning with the project cycle of February-March 2024. The workflow follows the structure outlined in the Gantt chart.

Expected Outcome

At the end of this project, we can expect to have solved some of the problems faced by the concerned business like if the data is categorised sales of all the products, this can help them be prepared for all the things and will be able to grow easily. Predicting sales using tools like Excel, Python with the help of data can help them predict better instead of relying on random assumption-based sales and going with that. They can get the optimised price and even stop making those products which are generating less revenue and start making more of those which generate more revenue.

